Exp No-2
DATE -

Study Of Network Cables

Aim: Study of different types of Network cables.

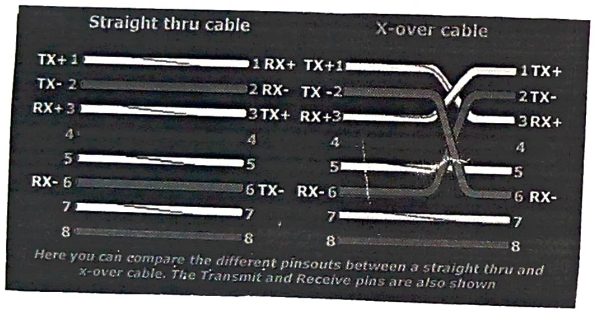
a) Understanding different types of network cable:

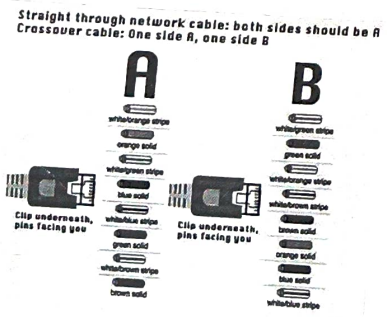
Different types of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable.

Cable type	Category	Maximum Data Transmission	Advantages/ Disadvantages	Application/Use
UTP	Category 3 Category 5 Category 5e	10bps upto 100Mbps 1Gbps	Advantages: • Cheaper in cost • Easy to install as they have a smaller overall diameter. Disadvantages: • More prone to (EMI) Electro magnetic interfere and noise. -Hae	10 Base-T Ethernet Fast Ethernet, Gigabit Ethernet Fast Ethernet, Gigabit Ethernet
STP	Category 6, 6a	10Gbps	Advantages: - Shielded - Faster than UTP - Less susceptible to noise and interference	Gigabit Ethernet, 10G Ethernet, (55m) widely used in data centres.
SSTP	Category 7	10Gbps	Disadvantages: Expensive Greater installation efforts	Gigabit Ethernet 10G Ethernet (100m)

Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	High bandwidth, Immune to interference Low loss bandwidth versatile Disadvantage Limited distance Cost Size is bulky	Speed of signal is 500 m Television network High speed internet connections
Fibre optics cable	Single mode Multi mode	100 Gbps	Advantages: High speed High bandwidth High security Long distance	Maximum distance of fibre optics cable is around 100 meters





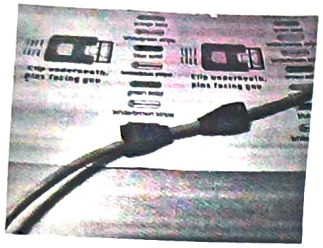
b) Make your own Ethernet Cross-over cable / straight cable.

Tools and parts needed

- Ethernet cabling, CAT5e is certified for gigabit support, but CAT 5 cabling works as well, just over shorter distances.
- A Crimping tool. This is an all in one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.
- Two RJ45 plugs
- ~~Optional~~ two plug shields.

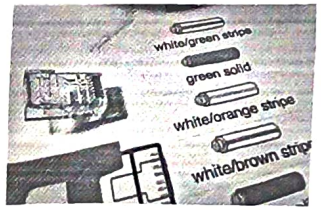
PROCEDURE:

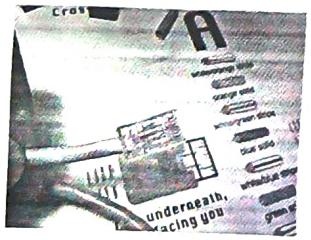
1. To start construction of the device, begin by threading shields onto the cable.
2. Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.











3. After you will need to untangle the wires, there should be four twisted pairs. Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.

4. Once the order is correct, bunch them together in a line, and if there any that stick ~~and~~ out out farther than others. Snip them back to even level. The difficult aspect is placing these into the RJ45 plug without messing up the order. To do so hold the plug with the clip side facing away from you and have the golden pins facing towards you.

5. Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shilding and if it isn't, that means you stripped off too much, shielding. Simply snip the cables back a little more.

6. After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

7. Lastly repeat for the other end using diagram B / diagram B.

RESULT: Thus, the different type of network cable has been studied and implemented successfully.